

MULTIS COMPLEX EP 2

SDS # : 30935

previous revision date : 2024/03/13

SECTION 1: Identification of the substance/mixture and of the company/undertaking

1.1 Product identifier

Product name : MULTIS COMPLEX EP 2

1.2 Relevant identified uses of the substance or mixture and uses advised against

Identified uses
Lubricating grease

1.3 Details of the supplier of the safety data sheet

TotalEnergies Lubrifiants
562 Avenue du Parc de L'île
92029 Nanterre Cedex FRANCE
Tél: +33 (0)1 41 35 40 00
Fax: +33 (0)1 41 35 84 71
rm.msds-lubs@totalenergies.com

TotalEnergies Marketing Deutschland GmbH
Jean-Monnet-Straße 2
10557 BERLIN
DEUTSCHLAND
Tel: +49 (0)30 2027 60

msds@totalenergies.com

Contact

HSE : + 49 (0) 30/ 2027-9429

1.4 Emergency telephone number

National advisory body/Poison Center

Telephone number : Giftnotruf Berlin, Tel.+49 (0)30 19240 (24 h erreichbar, Beratung in Deutsch und Englisch)

Supplier

Telephone number : TOTAL Emergency number: +49 89 220 61012

SECTION 2: Hazards identification

2.1 Classification of the substance or mixture

Product definition : Mixture

Classification according to Regulation (EC) No. 1272/2008 [CLP/GHS]

Not classified.

The product is not classified as hazardous according to Regulation (EC) 1272/2008 as amended.

For more details about adverse physical, human health and environmental effects, see sections 9 to 12.

2.2 Label elements

Signal word	: No signal word.
Hazard statements	:  No hazard statement.
<u>Precautionary statements</u>	
Prevention	: Not applicable.
Response	: Not applicable.
Storage	: Not applicable.
Disposal	: Not applicable.
Supplemental label elements	: Contains Naphthenic acids, zinc salts. May produce an allergic reaction. Safety data sheet available on request.
Annex XVII - Restrictions on the manufacture, placing on the market and use of certain dangerous substances, mixtures and articles	: Not applicable.

2.3 Other hazards

This mixture does not contain any substances that are assessed to be a PBT or a vPvB in a concentration $\geq 0,1$ %.

This product does not contain any substance present at a concentration equal to or greater than 0.1% by mass, included in the list drawn up in accordance with article 59, paragraph 1 of the REACH Regulation, due to its endocrine disrupting properties, or a substance known to have endocrine disrupting properties in accordance with the criteria set out in Commission Delegated Regulation (EU) 2017/2100 or Commission Regulation 2018/605.

Other hazards which do not result in classification : None known.

SECTION 3: Composition/information on ingredients

3.2 Mixtures : Mixture

Product/substance	Identifiers	% (w/w)	Classification	Specific Conc. Limits, M-factors and ATEs	Type
dilithium azelate	REACH #: 01-2120119814-57 EC: 254-184-4 CAS: 38900-29-7	≤ 3	Acute Tox. 4, H302	ATE [Oral] = 301 mg/kg	[1]
1-Propene, 2-methyl-, sulfurized	EC: 270-943-2 CAS: 68511-50-2	≤ 3	Aquatic Chronic 4, H413	-	[1]
Phosphorodithioic acid, mixed O,O-bis(2-ethylhexyl and iso-Bu) esters, zinc salts	REACH #: 01-2119948548-22 EC: 270-478-5 CAS: 68442-22-8	< 2.5	Skin Irrit. 2, H315 Eye Dam. 1, H318 Aquatic Chronic 2, H411	-	[1] [2]
methyl-1H-benzotriazole	REACH #: 01-2119979081-35 EC: 249-596-6 CAS: 29385-43-1	< 1	Acute Tox. 4, H302 Eye Irrit. 2, H319 Repr. 2, H361d Aquatic Chronic 2, H411	ATE [Oral] = 675 mg/kg	[1]
Naphthenic acids, zinc salts	REACH #: 01-2120783834-41 EC: 234-409-2 CAS: 12001-85-3	< 1	Eye Irrit. 2, H319 Skin Sens. 1B, H317 Aquatic Chronic 2, H411	-	[1] [2]

			See Section 16 for the full text of the H statements declared above.		
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Additional information : Mineral oil of petroleum origin. Product containing mineral oil with less than 3% DMSO extract as measured by IP 346

There are no additional ingredients present which, within the current knowledge of the supplier and in the concentrations applicable, are classified as hazardous to health or the environment, are PBTs, vPvBs or Substances of equivalent concern, or have been assigned a workplace exposure limit and hence require reporting in this section.

Type

[1] Substance classified with a health or environmental hazard

[2] Substance with a workplace exposure limit

Occupational exposure limits, if available, are listed in Section 8.

SECTION 4: First aid measures

4.1 Description of first aid measures

- Eye contact** : Immediately flush eyes with plenty of water, occasionally lifting the upper and lower eyelids. Check for and remove any contact lenses. Get medical attention if irritation occurs.
- Inhalation** : Remove victim to fresh air and keep at rest in a position comfortable for breathing. Get medical attention if symptoms occur.
- Skin contact** : Wash skin thoroughly with soap and water or use recognized skin cleanser. Remove contaminated clothing and shoes. Get medical attention if symptoms occur.
- Ingestion** : Wash out mouth with water. Do not induce vomiting unless directed to do so by medical personnel. Get medical attention if symptoms occur.
- Protection of first-aiders** : No action shall be taken involving any personal risk or without suitable training.

4.2 Most important symptoms and effects, both acute and delayed

- Eye contact** : No specific data.
- Inhalation** : No specific data.
- Skin contact** :  irritation
dryness
cracking
- Ingestion** : No specific data.

4.3 Indication of any immediate medical attention and special treatment needed

- Notes to physician** : Treat symptomatically. Contact poison treatment specialist immediately if large quantities have been ingested or inhaled.
- Specific treatments** : No specific treatment.

SECTION 5: Firefighting measures

5.1 Extinguishing media

- Suitable extinguishing media** : Use dry chemical, CO₂, water spray (fog) or foam.
- Unsuitable extinguishing media** : Do not use water jet.

5.2 Special hazards arising from the substance or mixture

Hazards from the substance or mixture : No specific fire or explosion hazard.

Hazardous combustion products : carbon monoxide
carbon dioxide
Silicon Dioxide
nitrogen oxides
phosphorus oxides
sulfur oxides
Hydrogen sulfide
Mercaptans
Zinc oxides

5.3 Advice for firefighters

Special protective actions for fire-fighters : Promptly isolate the scene by removing all persons from the vicinity of the incident if there is a fire. No action shall be taken involving any personal risk or without suitable training.

Special protective equipment for fire-fighters : Fire-fighters should wear appropriate protective equipment and self-contained breathing apparatus (SCBA) with a full face-piece operated in positive pressure mode. Clothing for fire-fighters (including helmets, protective boots and gloves) conforming to European standard EN 469 will provide a basic level of protection for chemical incidents.

SECTION 6: Accidental release measures

6.1 Personal precautions, protective equipment and emergency procedures

For non-emergency personnel : No action shall be taken involving any personal risk or without suitable training. Evacuate surrounding areas. Keep unnecessary and unprotected personnel from entering. Do not touch or walk through spilled material. Put on appropriate personal protective equipment.

For emergency responders : If specialized clothing is required to deal with the spillage, take note of any information in Section 8 on suitable and unsuitable materials. See also the information in "For non-emergency personnel".

6.2 Environmental precautions : Avoid dispersal of spilled material and runoff and contact with soil, waterways, drains and sewers. Inform the relevant authorities if the product has caused environmental pollution (sewers, waterways, soil or air).

6.3 Methods and materials for containment and cleaning up

Small spill : Move containers from spill area. Vacuum or sweep up material and place in a designated, labeled waste container. Dispose of via a licensed waste disposal contractor.

Large spill : Move containers from spill area. Prevent entry into sewers, water courses, basements or confined areas. Vacuum or sweep up material and place in a designated, labeled waste container. Dispose of via a licensed waste disposal contractor.

6.4 Reference to other sections : See Section 1 for emergency contact information.
See Section 8 for information on appropriate personal protective equipment.
See Section 13 for additional waste treatment information.

SECTION 7: Handling and storage

7.1 Precautions for safe handling

- Protective measures** : Put on appropriate personal protective equipment (see Section 8).
- Advice on general occupational hygiene** : Eating, drinking and smoking should be prohibited in areas where this material is handled, stored and processed. Workers should wash hands and face before eating, drinking and smoking. Remove contaminated clothing and protective equipment before entering eating areas. See also Section 8 for additional information on hygiene measures.

7.2 Conditions for safe storage, including any incompatibilities

Store in accordance with local regulations. Store in original container protected from direct sunlight in a dry, cool and well-ventilated area, away from incompatible materials (see Section 10) and food and drink. Keep container tightly closed and sealed until ready for use. Containers that have been opened must be carefully resealed and kept upright to prevent leakage. Do not store in unlabeled containers. Use appropriate containment to avoid environmental contamination. See Section 10 for incompatible materials before handling or use.

7.3 Specific end use(s)

- Recommendations** : Not available.
- Industrial sector specific solutions** : Not available.

SECTION 8: Exposure controls/personal protection

8.1 Control parameters

Occupational exposure limits

Product/substance	Exposure limit values
Phosphorodithioic acid, mixed O,O-bis (2-ethylhexyl and iso-Bu) esters, zinc salts	DFG MAK-values list (Germany, 7/2023) [Zinc and its inorganic compounds] Develop C. PEAK 15 minutes: 0.4 mg/m ³ 4 times per shift [Interval: 1 hour]. Form: respirable fraction. TWA 8 hours: 2 mg/m ³ . Form: inhalable fraction. TWA 8 hours: 0.1 mg/m ³ . Form: respirable fraction. PEAK 15 minutes: 4 mg/m ³ 4 times per shift [Interval: 1 hour]. Form: inhalable fraction.
Naphthenic acids, zinc salts	DFG MAK-values list (Germany, 7/2023) [Zinc and its inorganic compounds] Develop C. PEAK 15 minutes: 0.4 mg/m ³ 4 times per shift [Interval: 1 hour]. Form: respirable fraction. TWA 8 hours: 2 mg/m ³ . Form: inhalable fraction. TWA 8 hours: 0.1 mg/m ³ . Form: respirable fraction. PEAK 15 minutes: 4 mg/m ³ 4 times per shift [Interval: 1 hour]. Form: inhalable fraction.

Biological Limit Values (BLV)

No exposure indices known.

- Recommended monitoring procedures** : Reference should be made to monitoring standards, such as the following: European Standard EN 689 (Workplace atmospheres - Guidance for the assessment of exposure by inhalation to chemical agents for comparison with limit values and measurement strategy) European Standard EN 14042 (Workplace atmospheres - Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents) European Standard EN 482 (Workplace atmospheres - General requirements for the performance of procedures for the measurement of chemical agents) Reference to national guidance documents for methods for the determination of hazardous substances will also be

required.

Advisory OEL

: Mineral oil mist: USA: OSHA (PEL) TWA 5 mg/m³, NIOSH (REL) TWA 5 mg/m³, STEL 10 mg/m³, ACGIH (TLV) TWA 5 mg/m³ (highly refined)

DNELs/DMELs

Product/substance	Type	Exposure	Value	Population	Effects
dilithium azelate	DNEL	Long term Dermal	0.172 mg/cm ²	Workers	Local
	DNEL	Long term Dermal	0.023 mg/cm ²	General population	Local
Phosphorodithioic acid, mixed O,O-bis(2-ethylhexyl and iso-Bu) esters, zinc salts	DNEL	Long term Oral	0.24 mg/kg bw/day	General population	Systemic
	DNEL	Long term Inhalation	1.98 mg/m ³	General population	Systemic
	DNEL	Long term Dermal	5.71 mg/kg bw/day	General population	Systemic
	DNEL	Long term Inhalation	8.05 mg/m ³	Workers	Systemic
	DNEL	Long term Dermal	11.4 mg/kg bw/day	Workers	Systemic
methyl-1H-benzotriazole	DNEL	Long term Oral	0.01 mg/kg bw/day	General population	Systemic
	DNEL	Long term Dermal	0.01 mg/kg bw/day	General population	Systemic
	DNEL	Long term Dermal	0.3 mg/kg bw/day	Workers	Systemic
	DNEL	Long term Inhalation	350 µg/m ³	General population	Systemic
	DNEL	Long term Inhalation	21.2 mg/m ³	Workers	Systemic
Naphthenic acids, zinc salts	DNEL	Long term Oral	0.17 ng/kg bw/day	General population	Systemic
	DNEL	Long term Inhalation	0.29 mg/m ³	General population	Systemic
	DNEL	Long term Inhalation	1.18 mg/m ³	Workers	Systemic
	DNEL	Long term Dermal	1.7 mg/kg bw/day	General population	Systemic
	DNEL	Long term Dermal	3.3 mg/kg bw/day	Workers	Systemic

PNECs

Product/ingredient name	Compartment Detail	Name	Method Detail
dilithium azelate	Fresh water	0.023 mg/l	-
	Marine water	0.0023 mg/l	-
Phosphorodithioic acid, mixed O,O-bis(2-ethylhexyl and iso-Bu) esters, zinc salts	Fresh water	0.004 mg/l	-
	Marine water	0.0046 mg/l	-
	Fresh water sediment	0.04508 mg/kg dwt	-
	Marine water sediment	0.005 mg/kg dwt	-
	Soil	0.007 mg/kg dwt	-
	Sewage Treatment Plant	100 mg/l	-
	Secondary Poisoning	10.67 mg/kg	-
methyl-1H-benzotriazole	Fresh water	8 µg/l	-
	Marine water	20 µg/l	-
	Sewage Treatment Plant	39.4 mg/l	-

Naphthenic acids, zinc salts	Fresh water sediment	117 µg/kg dwt	-
	Marine water sediment	292 µg/kg dwt	-
	Soil	18.7 µg/kg dwt	-
	Fresh water sediment	15.1 to 19438.4 µg/kg dwt	-
	Marine water sediment	1.51 to 1943.84 µg/kg dwt	-
	Sewage Treatment Plant	689.7 µg/l	-
	Fresh water	4 µg/l	-
	Marine water	400 ng/l	-

8.2 Exposure controls

Appropriate engineering controls : Good general ventilation should be sufficient to control worker exposure to airborne contaminants.

Individual protection measures

Hygiene measures : Wash hands, forearms and face thoroughly after handling chemical products, before eating, smoking and using the lavatory and at the end of the working period. Appropriate techniques should be used to remove potentially contaminated clothing. Wash contaminated clothing before reusing. Ensure that eyewash stations and safety showers are close to the workstation location.

Eye/face protection : In case of contact through splashing: safety glasses with side-shields, EN 166.

Skin protection

Hand protection : Chemical-resistant, impervious gloves complying with an approved standard should be worn at all times when handling chemical products if a risk assessment indicates this is necessary.
Hydrocarbon-proof gloves
nitrile rubber
Fluorinated rubber
Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. Also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion, and the contact time.
In case of prolonged contact with the product, it is recommended to wear gloves complying with ISO 21420 and EN 374 standards, protecting at least for 480 minutes and having a thickness of 0,38 mm at least. These values are indicative only. The level of protection is provided by the material of the glove, its technical characteristics, its resistance to the chemicals to be handled, the appropriateness of its use and its replacement frequency

Body protection :  Personal protective equipment for the body should be selected based on the task being performed and the risks involved and should be approved by a specialist before handling this product.

Respiratory protection : None under normal use conditions. If these are not sufficient to maintain exposure below the OEL, suitable respiratory protection must be worn (Type A/P1).

Environmental exposure controls : Emissions from ventilation or work process equipment should be checked to ensure they comply with the requirements of environmental protection legislation. In some cases, fume scrubbers, filters or engineering modifications to the process equipment will be necessary to reduce emissions to acceptable levels.

SECTION 9: Physical and chemical properties

The conditions of measurement of all properties are at standard temperature (20°C / 68°F) and pressure (1013 hPa) unless otherwise indicated

9.1 Information on basic physical and chemical properties

Appearance

Physical state	: Solid. [grease]	
Color	: Red.	
Odor	: Not available.	
pH	: Not applicable.	Product is non-soluble (in water).
Melting point/freezing point	: Not applicable.	
Initial boiling point and boiling range	: Technically not possible to measure	
Flash point	: Not applicable.	
Flammability	: Yes.	
Lower and upper explosion limit	: Not applicable.	
Vapor pressure	: Not applicable.	
Vapor density	: Not applicable.	
Relative density	: 0.9 [ISO EN 3675]	
Density	: 0.9 g/cm ³ [20°C]	
Solubility(ies)	:	

Media	Result
water	Not soluble

Miscible with water	: No.
Partition coefficient: n-octanol/ water	: >3.5
Auto-ignition temperature	: Not applicable.
Decomposition temperature	: >260°C
Viscosity	: ☒ Dynamic (room temperature): Not available. Kinematic (room temperature): Not available. Kinematic (40°C): Not applicable.

Particle characteristics

Median particle size	: Not available.
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9.2 Other information

No other relevant physical and chemical parameters for the safe use of the product

SECTION 10: Stability and reactivity

10.1 Reactivity	: No specific test data related to reactivity available for this product or its ingredients.
10.2 Chemical stability	: Stable under recommended storage and handling conditions (see Section 7).
10.3 Possibility of hazardous reactions	: Under normal conditions of storage and use, hazardous reactions will not occur.

10.4 Conditions to avoid : Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking.

10.5 Incompatible materials : Strong oxidizing agents

10.6 Hazardous decomposition products : Under normal conditions of storage and use, hazardous decomposition products should not be produced.

SECTION 11: Toxicological information

11.1 Information on hazard classes as defined in Regulation (EC) No 1272/2008

Acute toxicity

Product/substance	Result	Species	Dose	Exposure	Test
dilithium azelate	LD50 Dermal	Rat	>2000 mg/kg	-	-
	LD50 Oral	Rat	301 mg/kg	-	-
1-Propene, 2-methyl-, sulfurized	LD50 Oral	Rat	8.6 g/kg	-	-
Phosphorodithioic acid, mixed O,O-bis(2-ethylhexyl and iso-Bu) esters, zinc salts	LD50 Dermal	Rabbit	>2000 mg/kg	-	OECD 402 Acute Dermal Toxicity
	LD50 Oral	Rat	>2000 mg/kg	-	EPA
methyl-1H-benzotriazole	LD50 Dermal	Rabbit - Male, Female	>2000 mg/kg	-	OECD 402
	LD50 Oral	Rat	675 mg/kg	-	-
	LD50 Oral	Rat - Male, Female	720 mg/kg	-	OECD 401
Naphthenic acids, zinc salts	LD50 Dermal	Rabbit	2500 mg/kg	-	-
	LD50 Oral	Mouse - Female	>2000 mg/kg	-	OECD 401
	LD50 Oral	Rat	4920 mg/kg	-	-

Acute toxicity estimates

Product/substance	Oral (mg/kg)	Dermal (mg/kg)	Inhalation (gases) (ppm)	Inhalation (vapors) (mg/l)	Inhalation (dusts and mists) (mg/l)
MULTIS COMPLEX EP 2	15759.2	N/A	N/A	N/A	N/A
dilithium azelate	301	N/A	N/A	N/A	N/A
1-Propene, 2-methyl-, sulfurized	8600	N/A	N/A	N/A	N/A
methyl-1H-benzotriazole	675	N/A	N/A	N/A	N/A
Naphthenic acids, zinc salts	4920	2500	N/A	N/A	N/A

Conclusion/Summary : Based on available data, the classification criteria are not met.

Irritation/Corrosion

Product/substance	Result	Species	Score	Exposure	Test
Phosphorodithioic acid, mixed O,O-bis(2-ethylhexyl and iso-Bu) esters, zinc salts	Eyes - Severe irritant	Rabbit	-	-	-
	Skin - Irritant	Rabbit	-	-	OECD 404 Acute Dermal Irritation/Corrosion
methyl-1H-benzotriazole	Eyes - Iris lesion	Rabbit	0	-	OECD 405



Naphthenic acids, zinc salts	Eyes - Mild irritant	Rabbit	-	10 mg	-
	Skin - Edema	Rabbit	0	4 hours	OECD 404
	Skin - Mild irritant	Rabbit	-	0.5 MI	-
	Skin - Mild irritant	Rabbit	-	24 hours 500 mg	-

Conclusion/Summary

- Skin** : Based on available data, the classification criteria are not met.
- Eyes** : The supplier of one or more of the components contained within this formulation has indicated that he has data on the components and/or similar mixtures, which confirms that at the concentration used, classification is not required.
- Respiratory** : Based on available data, the classification criteria are not met.

Sensitization

Product/substance	Route of exposure	Species	Result
Phosphorodithioic acid, mixed O,O-bis(2-ethylhexyl and iso-Bu) esters, zinc salts	skin	Guinea pig	Not sensitizing
methyl-1H-benzotriazole	skin	Guinea pig	Not sensitizing
Naphthenic acids, zinc salts	skin	Human	Sensitizing

Conclusion/Summary

- Skin** : Based on available data, the classification criteria are not met. Contains sensitizer. May produce an allergic reaction.
- Respiratory** : Based on available data, the classification criteria are not met.

Mutagenicity

Product/substance	Test	Experiment	Result
Phosphorodithioic acid, mixed O,O-bis(2-ethylhexyl and iso-Bu) esters, zinc salts	OECD 471 Bacterial Reverse Mutation Test	Experiment: In vitro Subject: Bacteria Cell: Somatic	Negative
	OECD 476 <i>In vitro</i> Mammalian Cell Gene Mutation Test	Experiment: In vitro Subject: Mammalian-Animal	Negative
	OECD 474 Mammalian Erythrocyte Micronucleus Test	Experiment: In vivo Subject: Mammalian-Animal Cell: Somatic	Negative
	OECD 471	Experiment: In vitro Subject: Bacteria	Negative
methyl-1H-benzotriazole	OECD 476 Read across	Experiment: In vitro Subject: Mammalian-Animal	Negative
	OECD 474	Experiment: In vivo Subject: Mammalian-Animal	Negative

- Conclusion/Summary** : Based on available data, the classification criteria are not met.

Carcinogenicity

- Conclusion/Summary** : Based on available data, the classification criteria are not met.

Reproductive toxicity

Product/substance	Maternal toxicity	Fertility	Development toxin	Species	Dose	Exposure
Phosphorodithioic acid, mixed O,O-bis(2-ethylhexyl and iso-Bu) esters, zinc salts	Negative	Negative	Negative	Rat	Oral: 160 mg/kg NOAEL	-

- Conclusion/Summary** : Based on available data, the classification criteria are not met.

Teratogenicity



TotalEnergies

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Product/substance	Result	Species	Dose	Exposure
methyl-1H-benzotriazole	Negative - Oral	Rat - Female	-	-

Conclusion/Summary : Based on available data, the classification criteria are not met.

Specific target organ toxicity (single exposure)

Conclusion/Summary : Based on available data, the classification criteria are not met.

Specific target organ toxicity (repeated exposure)

Conclusion/Summary : Based on available data, the classification criteria are not met.

Aspiration hazard

Conclusion/Summary : Based on available data, the classification criteria are not met.

Information on the likely routes of exposure : Not available.

Potential acute health effects

Eye contact : No known significant effects or critical hazards.
Inhalation : No known significant effects or critical hazards.
Skin contact : Defatting to the skin. May cause skin dryness and irritation.
Ingestion : No known significant effects or critical hazards.

Symptoms related to the physical, chemical and toxicological characteristics

Eye contact : No specific data.
Inhalation : No specific data.
Skin contact : 
irritation
dryness
cracking
Ingestion : No specific data.

Delayed and immediate effects and also chronic effects from short and long term exposure

Short term exposure

Potential immediate effects : Not available.
Potential delayed effects : Not available.

Long term exposure

Potential immediate effects : Not available.
Potential delayed effects : Not available.

Potential chronic health effects

Product/substance	Result	Species	Dose	Exposure
Phosphorodithioic acid, mixed O,O-bis(2-ethylhexyl and iso-Bu) esters, zinc salts	Sub-acute NOAEL Oral	Rat	160 mg/kg	-
methyl-1H-benzotriazole	Sub-chronic NOAEL Oral	Rat - Male, Female	150 mg/kg	28 days; 7 days per week

Conclusion/Summary : Not available.
General : No known significant effects or critical hazards.
Carcinogenicity : No known significant effects or critical hazards.
Mutagenicity : No known significant effects or critical hazards.

Reproductive toxicity : No known significant effects or critical hazards.

11.2 Information on other hazards

11.2.1 Endocrine disrupting properties

This product does not contain any substance present at a concentration equal to or greater than 0.1% by mass, included in the list drawn up in accordance with article 59, paragraph 1 of the REACH Regulation, due to its endocrine disrupting properties, or a substance known to have endocrine disrupting properties in accordance with the criteria set out in Commission Delegated Regulation (EU) 2017/2100 or Commission Regulation 2018/605.

11.2.2 Other information

Not available.

SECTION 12: Ecological information

12.1 Toxicity

Product/substance	Result	Species	Exposure	Test
dilithium azelate	Acute LC50 >100 mg/l	Algae	72 hours	-
1-Propene, 2-methyl-, sulfurized	Acute LC50 >100 mg/l	Daphnia	48 hours	-
	Acute EC50 >100 mg/l	Algae	72 hours	-
Phosphorodithioic acid, mixed O,O-bis(2-ethylhexyl and iso-Bu) esters, zinc salts	Acute EC50 1000 mg/l	Daphnia	48 hours	-
	Acute LC50 1000 mg/l	Fish	96 hours	-
	Acute EC50 24 mg/l	Algae - <i>Scenedesmus subspicatus</i>	72 hours	OECD 201
	Acute EC50 23 mg/l	Daphnia - <i>Daphnia Magna</i>	48 hours	OECD 202
	Acute LC50 4.5 mg/l	Fish	96 hours	-
	Acute NOEC 0.4 mg/l	Daphnia - <i>Daphnia Magna</i>	21 days	-
methyl-1H-benzotriazole	Acute EC10 1.18 mg/l	Algae - <i>Desmodesmus subspicatus</i>	72 hours	OECD 201
	Fresh water	Daphnia	48 hours	OECD 202
	Acute EC50 8.58 mg/l			
	Fresh water			
	Acute LC50 55 mg/l Marine water	Fish - <i>Cyprinodon variegatus</i>	96 hours	OECD 203
	Acute NOEC 30 mg/l	Fish - <i>Cyprinodon variegatus</i>	96 hours	OECD 203
	Marine water			
	Chronic NOEC 0.4 mg/l	Daphnia	21 days	OECD 211
	Fresh water			
Naphthenic acids, zinc salts	Acute EC50 4 mg/l	Algae - <i>Pseudokirchneriella subcapitata</i>	72 hours	OECD 201
	Acute EC50 100 mg/l	Daphnia - <i>Daphnia magna</i>	48 hours	-
	Acute LC50 92 ppm Fresh water	Fish - <i>Lepomis macrochirus</i>	96 hours	US EPA
	Chronic NOEC 1 mg/l	Algae - <i>Pseudokirchneriella subcapitata</i>	72 hours	OECD 201

Conclusion/Summary : Not available.

12.2 Persistence and degradability

Conclusion/Summary : Not available.



Product/substance	Aquatic half-life	Photolysis	Biodegradability
Phosphorodithioic acid, mixed O,O-bis(2-ethylhexyl and iso-Bu) esters, zinc salts	-	-	Not readily
methyl-1H-benzotriazole	-	-	Not readily
Naphthenic acids, zinc salts	-	-	Not readily

12.3 Bioaccumulative potential

Product/substance	LogK _{ow}	BCF	Potential
MULTIS COMPLEX EP 2	>3.5	-	Low
Phosphorodithioic acid, mixed O,O-bis(2-ethylhexyl and iso-Bu) esters, zinc salts	1.67	-	Low
methyl-1H-benzotriazole	1.081	-	Low
Naphthenic acids, zinc salts	1.89 to 11.15	-	High

12.4 Mobility in soil

Soil/water partition coefficient (K_{oc}) : Not available.

Mobility : Not available.

Mobility in soil : Given its physical and chemical characteristics, the product has no soil mobility. The product is insoluble and floats on water. Loss by evaporation is limited.

12.5 Results of PBT and vPvB assessment

This mixture does not contain any substances that are assessed to be a PBT or a vPvB in a concentration $\geq 0,1\%$.

12.6 Endocrine disrupting properties

This product does not contain any substance present at a concentration equal to or greater than 0.1% by mass, included in the list drawn up in accordance with article 59, paragraph 1 of the REACH Regulation, due to its endocrine disrupting properties, or a substance known to have endocrine disrupting properties in accordance with the criteria set out in Commission Delegated Regulation (EU) 2017/2100 or Commission Regulation 2018/605.

12.7 Other adverse effects

No known significant effects or critical hazards.

SECTION 13: Disposal considerations

13.1 Waste treatment methods

Product

Methods of disposal : The generation of waste should be avoided or minimized wherever possible. Disposal of this product, solutions and any by-products should at all times comply with the requirements of environmental protection and waste disposal legislation and any regional local authority requirements. Dispose of surplus and non-recyclable products via a licensed waste disposal contractor. Waste should not be disposed of untreated to the sewer unless fully compliant with the requirements of all authorities with jurisdiction.

Hazardous waste : Yes.
According to the European Waste Catalogue, Waste Codes are not product specific, but application specific. Waste codes should be assigned by the user based on the application for which the product was used. The following Waste Codes are only suggestions: 12 01 12*

Packaging

- Methods of disposal** : The generation of waste should be avoided or minimized wherever possible. Waste packaging should be recycled. Incineration or landfill should only be considered when recycling is not feasible.
- Special precautions** : This material and its container must be disposed of in a safe way. Empty containers or liners may retain some product residues. Avoid dispersal of spilled material and runoff and contact with soil, waterways, drains and sewers.

SECTION 14: Transport information

	ADR/RID	ADN	IMDG	ICAO/IATA
14.1 UN number or ID number	Not regulated.	Not regulated.	Not regulated.	Not regulated.
14.2 UN proper shipping name	-	-	-	-
14.3 Transport hazard class(es)	-	-	-	-
14.4 Packing group	-	-	-	-
14.5 Environmental hazards	No.	No.	No.	No.

- 14.6 Special precautions for user** : **Transport within user's premises:** always transport in closed containers that are upright and secure. Ensure that persons transporting the product know what to do in the event of an accident or spillage.

- 14.7 Maritime transport in bulk according to IMO instruments** : Not available.

SECTION 15: Regulatory information

15.1 Safety, health and environmental regulations/legislation specific for the substance or mixture

EU Regulation (EC) No. 1907/2006 (REACH)

Annex XIV - List of substances subject to authorization

Annex XIV

None of the components are listed.

Substances of very high concern

None of the components are listed.

Annex XVII - Restrictions on the manufacture, placing on the market and use of certain dangerous substances, mixtures and articles

Labeling : Not applicable.

Other EU regulations

Take note of Directive 98/24/EC on the protection of the health and safety of workers from the risks related to chemical agents at work

Industrial emissions (integrated pollution prevention and control) - Air : Not listed



Industrial emissions : Not listed
(integrated pollution prevention and control) -
Water

Explosive precursors : Not applicable.

Ozone depleting substances (1005/2009/EU)

Not listed.

Prior Informed Consent (PIC) (649/2012/EU)

Not listed.

Persistent Organic Pollutants

Not listed.

Seveso Directive

This product is not controlled under the Seveso Directive.

National regulations

Product/ingredient name	List name	Name on list	Classification	Notes
Phosphorodithioic acid, mixed O,O-bis(2-ethylhexyl and iso-Bu) esters, zinc salts	DFG MAK-values list	Zinc and its inorganic compounds	Develop C	-
Naphthenic acids, zinc salts	DFG MAK-values list	Zinc and its inorganic compounds	Develop C	-

Storage class (TRGS 510) : 11

Hazardous incident ordinance

This product is not controlled under the Germany Hazardous Incident Ordinance.

Hazard class for water : 1

Technical instruction on air quality control : ☒ Number 5.2.1: 12.09%
Number 5.2.5: 79.8200000%
Number 5.2.5 - Class I: 0.7410153%

National regulations : Waste Oil Ordinance (AltöIV) §7: This oil should be taken to a waste oil collection point after use. Improper disposal of waste oil harms the environment! Admixture of foreign substances such as solvents, brake fluids and coolants prohibited

International regulations

Chemical Weapon Convention List Schedules I, II & III Chemicals

Not listed.

Montreal Protocol

Not listed.

Stockholm Convention on Persistent Organic Pollutants

Not listed.

Rotterdam Convention on Prior Informed Consent (PIC)

Not listed.

UNECE Aarhus Protocol on POPs and Heavy Metals

Not listed.

LU - Luxembourg prohibited chemicals in the workplace

Not listed.

Inventory list

Australia inventory (AIIIC)	: All components are listed or exempted.
Canada inventory (DSL/NDSL)	: All components are listed or exempted.
China inventory (IECSC)	: All components are listed or exempted.
Europe inventory (EC)	: All components are listed or exempted.
Japan inventory	: Japan inventory (CSCL) : All components are listed or exempted. Japan inventory (ISHL) : Not determined.
New Zealand Inventory of Chemicals (NZIoC)	: At least one component is not listed.
Philippines inventory (PICCS)	: All components are listed or exempted.
Korea inventory (KECI)	: All components are listed or exempted.
Taiwan Chemical Substances Inventory (TCSI)	: All components are listed or exempted.
Thailand inventory	: Not determined.
Turkey inventory	: Not determined.
United States inventory (TSCA 8b)	: All components are listed or exempted.
Vietnam inventory	: Not determined.

The information stated in this section relates solely to the conformity of the chemical product with the countries Inventories. The information used to confirm the inventory status of this product may be based on additional data to the chemical composition shown in Section 3. Other regulations may apply for importation or marketing authorizations.

15.2 Chemical Safety Assessment : Risk management measures and safety conditions of use are included in the relevant sections of the SDS

SECTION 16: Other information

Indicates information that has changed from previously issued version.

Abbreviations and acronyms	: ACGIH = American Conference of Governmental Industrial Hygienists ATE = Acute Toxicity Estimate BCF = Bioconcentration Factor CLP = Classification, Labelling and Packaging Regulation [Regulation (EC) No. 1272/2008] DNEL = Derived No Effect Level DMEL = Derived Minimal Effect Level DMSO = Dimethyl Sulfoxide EL50 = median Effective Loading EUH statement = CLP-specific Hazard statement HSE = Health, Safety and Environment IC50 = Half maximal inhibitory concentration IDHL = Immediately dangerous to life or health LC50 = Median lethal concentration LD50 = Median lethal dose LL50 = median Lethal Loading LogKow = logarithm of the octanol/water partition coefficient N/A = Not available NIOSH = National Institute of Occupational Safety and Health NOAEL = No Observed Adverse Effect Level NOEC = No Observed Effect Concentration NOEL = No Observed Effect Level NOELR = No observed Effect Loading Rate OECD = Organisation for Economic Co-operation and Development
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OEL = Occupational Exposure Limit
 PBT = Persistent, Bioaccumulative and Toxic
 PNEC = Predicted No Effect Concentration
 QSAR = Quantitative Structure–Activity Relationship
 REL = Recommended Exposure Limit
 STEL = Short Term Exposure Limit
 TLV = Threshold Limit Value
 TWA = Time Weight Average
 VOC = Volatile Organic Compound
 vPvB = Very Persistent and Very Bioaccumulative
 Unique Formula Identifier (UFI)
 UVCB Substance of unknown or Variable composition, Complex reaction products or Biological material

Procedure used to derive the classification according to Regulation (EC) No. 1272/2008 [CLP/GHS]

Classification	Justification
Not classified.	

Full text of abbreviated H statements

H302 H315 H317 H318 H319 H361d H411 H413	Harmful if swallowed. Causes skin irritation. May cause an allergic skin reaction. Causes serious eye damage. Causes serious eye irritation. Suspected of damaging the unborn child. Toxic to aquatic life with long lasting effects. May cause long lasting harmful effects to aquatic life.
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Full text of classifications [CLP/GHS]

Acute Tox. 4 Aquatic Chronic 2 Aquatic Chronic 4 Eye Dam. 1 Eye Irrit. 2 Repr. 2 Skin Irrit. 2 Skin Sens. 1B	ACUTE TOXICITY - Category 4 AQUATIC HAZARD (LONG-TERM) - Category 2 AQUATIC HAZARD (LONG-TERM) - Category 4 SERIOUS EYE DAMAGE/ EYE IRRITATION - Category 1 SERIOUS EYE DAMAGE/ EYE IRRITATION - Category 2 TOXIC TO REPRODUCTION - Category 2 SKIN CORROSION/IRRITATION - Category 2 SKIN SENSITIZATION - Category 1B
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Date of revision : 2024/08/14
 previous revision date : 2024/03/13



TotalEnergies

MULTIS COMPLEX EP 2

SDS # : 30935

Version : 3.02

Notice to reader

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Final determination of suitability of any material is the sole responsibility of the user. All materials may present unknown hazards and should be used with caution. Although certain hazards are described herein, we cannot guarantee that these are the only hazards that exist.